In [42]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import duckdb

from sklearn.preprocessing import LabelEncoder
from sklearn.preprocessing import StandardScaler
from imblearn.over_sampling import SMOTE #Help with class imbalance for a classifie

from sklearn.decomposition import PCA
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report


from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier
from sklearn.linear_model import LogisticRegression

import pickle
```

In [22]:
```
sql_departments = duckdb.sql("SELECT * FROM 'departments.csv'")
sql_encounters = duckdb.sql("SELECT * FROM 'encounters.csv'")
sql_patients = duckdb.sql("SELECT * FROM 'patients.csv'")
sql_diagnosis = duckdb.sql("SELECT * FROM 'diagnosis.csv'")
sql_providers = duckdb.sql("SELECT * FROM 'providers.csv'")
sql_social_determinant = duckdb.sql("SELECT * FROM 'social_determinants.csv'")
sql_tiger = duckdb.sql("SELECT * FROM 'tigercensuscodes.csv'")
```

In [23]:
```
combined_data = duckdb.sql("""
    WITH duped_diagnosis AS (
        SELECT *
        FROM (
            SELECT *,
                ROW_NUMBER() OVER (PARTITION BY DiagnosisKey ORDER BY DiagnosisKey)
            FROM sql_diagnosis
        )
        WHERE rn = 1
    ),
    duped_providers AS (
        SELECT *
        FROM (
            SELECT *,
                ROW_NUMBER() OVER (PARTITION BY DurableKey, Type ORDER BY DurableKe
            FROM sql_providers
        )
        WHERE rn = 1
    ),
    duped_patients AS (
        SELECT *
        FROM (
            SELECT *,
                ROW_NUMBER() OVER (PARTITION BY DurableKey ORDER BY DurableKey) AS
```

```
            FROM sql_patients
        )
        WHERE rn = 1
    ),
    duped_departments AS (
        SELECT *
        FROM (
            SELECT *,
                ROW_NUMBER() OVER (PARTITION BY DepartmentKey ORDER BY DepartmentKe
            FROM sql_departments
        )
        WHERE rn = 1
    )
    SELECT
        e.*,
        p.CensusBlockGroupFipsCode, p.FirstRace, p.MaritalStatus, p.MyChartStatus,
        p.OmbEthnicity, p.OmbRace, p.SexAssignedAtBirth, p.SexualOrientation,
        p.SmokingStatus, p.VitalStatus, p.PatientBirthYearBin,
        d.GroupName, d.GroupCode, d.DiagnosisName, d.DiagnosisValue,
        pr.ClinicianTitle, pr.OfficeAddress, pr.OfficeCity, pr.OfficePostalCode,
        pr.PrimaryDepartment, pr.PrimarySpecialty,
        de.Address, de.City, de.County, de.DepartmentName,
        de.DepartmentSpecialty, de.DepartmentType, de.PostalCode, de.CensusTract
    FROM sql_encounters AS e
    LEFT JOIN duped_patients AS p
        ON e.PatientDurableKey = p.DurableKey
    LEFT JOIN duped_diagnosis AS d
        ON e.PrimaryDiagnosisKey = d.DiagnosisKey
    LEFT JOIN duped_providers AS pr
        ON e.ProviderDurableKey = pr.DurableKey AND e.Type = pr.Type
    LEFT JOIN duped_departments AS de
        ON e.DepartmentKey = de.DepartmentKey
""")

duckdb.sql("SELECT COUNT(*) FROM combined_data")
```

```
FloatProgress(value=0.0, layout=Layout(width='auto'), style=ProgressStyle(bar_color
='black'))
```

Out[23]:

```
┌──────────────┐
│ count_star() │
│    int64     │
├──────────────┤
│      7675801 │
└──────────────┘
```

Looking at Emergency Admissions

In [24]:

```
pd.set_option('display.max_colwidth', None)
duckdb.sql("""
    SELECT GroupName, COUNT(*) as Count
    FROM combined_data
    WHERE IsEdVisit = '1'
    GROUP BY GroupName
    ORDER BY Count DESC
""").df().head(10)
```

```
FloatProgress(value=0.0, layout=Layout(width='auto'), style=ProgressStyle(bar_color
='black'))
```

Out[24]:

	GroupName	Count
0	Pain in throat and chest	12023
1	Abdominal and pelvic pain	11956
2	Persons encountering health services for specific procedures and treatment, not carried out	7376
3	Dorsalgia	7070
4	Other sepsis	6838
5	Other joint disorder, not elsewhere classified	6230
6	Nausea and vomiting	4551
7	Other and unspecified soft tissue disorders, not elsewhere classified	4259
8	COVID-19	3711
9	Open wound of head	3599

In [73]:
```
emergency_df = duckdb.sql("""
    SELECT *
    FROM combined_data
    WHERE IsEdVisit = '1'
""").df()
emergency_df.shape[0]
```

```
FloatProgress(value=0.0, layout=Layout(width='auto'), style=ProgressStyle(bar_color
='black'))
```

Out[73]: 261458

In [27]:
```
emergency_df = emergency_df[emergency_df["GroupName"].isin(["Abdominal and pelvic p
emergency_df.info()
```

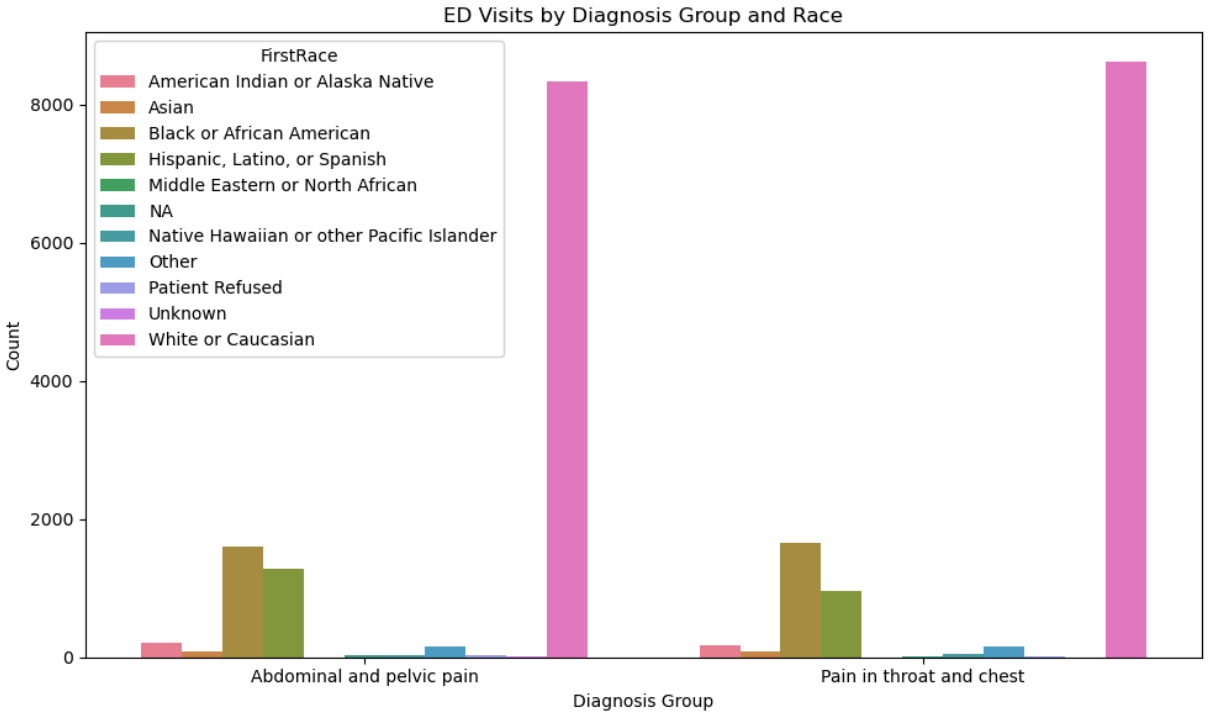
```
<class 'pandas.core.frame.DataFrame'>
Index: 23979 entries, 12 to 261383
Data columns (total 60 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   Date                          23979 non-null  datetime64[us]
 1   AdmissionInstant              23979 non-null  object
 2   AdmitYear                     23979 non-null  object
 3   AdmitMonth                    23979 non-null  object
 4   AdmitDay                      23979 non-null  object
 5   AdmitHour                     23979 non-null  object
 6   AdmitMinute                   23979 non-null  object
 7   AdmissionSource               23976 non-null  object
 8   AdmissionType                 23979 non-null  object
 9   DischargeInstant              23979 non-null  object
 10  DischargeYear                 23979 non-null  object
 11  DischargeMonth                23979 non-null  object
 12  DischargeDay                  23979 non-null  object
 13  DischargeHour                 23979 non-null  object
 14  DischargeMinute               23979 non-null  object
 15  EncounterKey                  23979 non-null  int64
 16  PatientDurableKey             23979 non-null  int64
 17  Type                          23979 non-null  object
 18  VisitType                     23979 non-null  object
 19  VisitTypeDescription          23979 non-null  object
 20  ProviderDurableKey            23979 non-null  int64
 21  AttendingProviderDurableKey   23979 non-null  int64
 22  DischargeProviderDurableKey   23979 non-null  int64
 23  DepartmentKey                 23979 non-null  int64
 24  PrimaryDiagnosisKey           23979 non-null  int64
 25  IsEdVisit                     23979 non-null  int64
 26  IsHospitalAdmission           23979 non-null  int64
 27  IsHospitalOutpatientVisit     23979 non-null  int64
 28  IsInpatientAdmission          23979 non-null  int64
 29  IsObservation                 23979 non-null  int64
 30  IsOutpatientFaceToFaceVisit   23979 non-null  int64
 31  CensusBlockGroupFipsCode      23445 non-null  object
 32  FirstRace                     23445 non-null  object
 33  MaritalStatus                 23445 non-null  object
 34  MyChartStatus                 23445 non-null  object
 35  OmbEthnicity                  23445 non-null  object
 36  OmbRace                       23445 non-null  object
 37  SexAssignedAtBirth            23445 non-null  object
 38  SexualOrientation             23445 non-null  object
 39  SmokingStatus                 23445 non-null  object
 40  VitalStatus                   23445 non-null  object
 41  PatientBirthYearBin           23445 non-null  object
 42  GroupName                     23979 non-null  object
 43  GroupCode                     23979 non-null  object
 44  DiagnosisName                 23979 non-null  object
 45  DiagnosisValue                23979 non-null  object
 46  ClinicianTitle                0 non-null      object
 47  OfficeAddress                 0 non-null      object
 48  OfficeCity                    0 non-null      object
 49  OfficePostalCode              0 non-null      object
 50  PrimaryDepartment             0 non-null      object
```

```
 51  PrimarySpecialty            0 non-null      object
 52  Address                     23979 non-null  object
 53  City                        23979 non-null  object
 54  County                      23979 non-null  object
 55  DepartmentName              23979 non-null  object
 56  DepartmentSpecialty         23979 non-null  object
 57  DepartmentType              23979 non-null  object
 58  PostalCode                  23979 non-null  object
 59  CensusTract                 23979 non-null  object
dtypes: datetime64[us](1), int64(13), object(46)
memory usage: 11.2+ MB
```

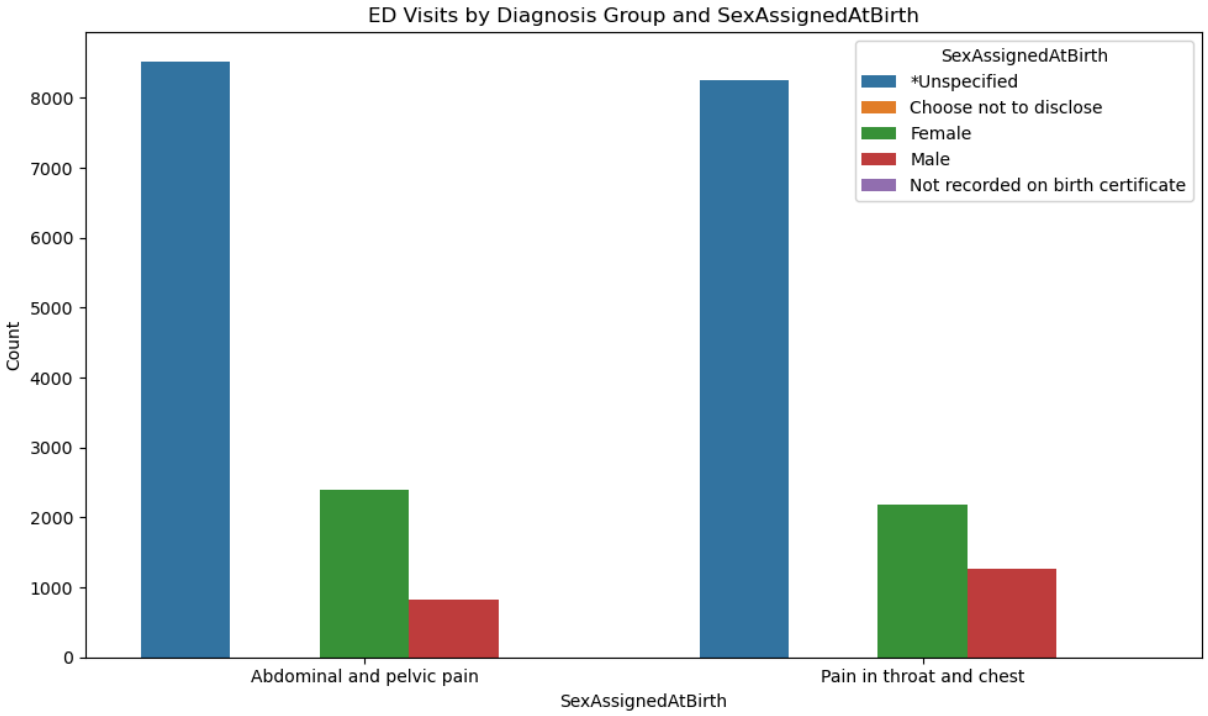
In [76]:
```
drop_cols = [
    # All null
    'ClinicianTitle', 'OfficeAddress', 'OfficeCity', 'OfficePostalCode',
    'PrimaryDepartment', 'PrimarySpecialty',
    # Redundant time
    'AdmissionInstant', 'DischargeInstant',
    'AdmitYear', 'AdmitMonth', 'AdmitDay', 'AdmitHour', 'AdmitMinute',
    'DischargeYear', 'DischargeMonth', 'DischargeDay', 'DischargeHour', 'DischargeM
    # Redundant keys
    'AttendingProviderDurableKey', 'DischargeProviderDurableKey'
]

emergency_df = emergency_df.drop(columns=drop_cols)
print(emergency_df.shape)
```

```
(261458, 40)
```

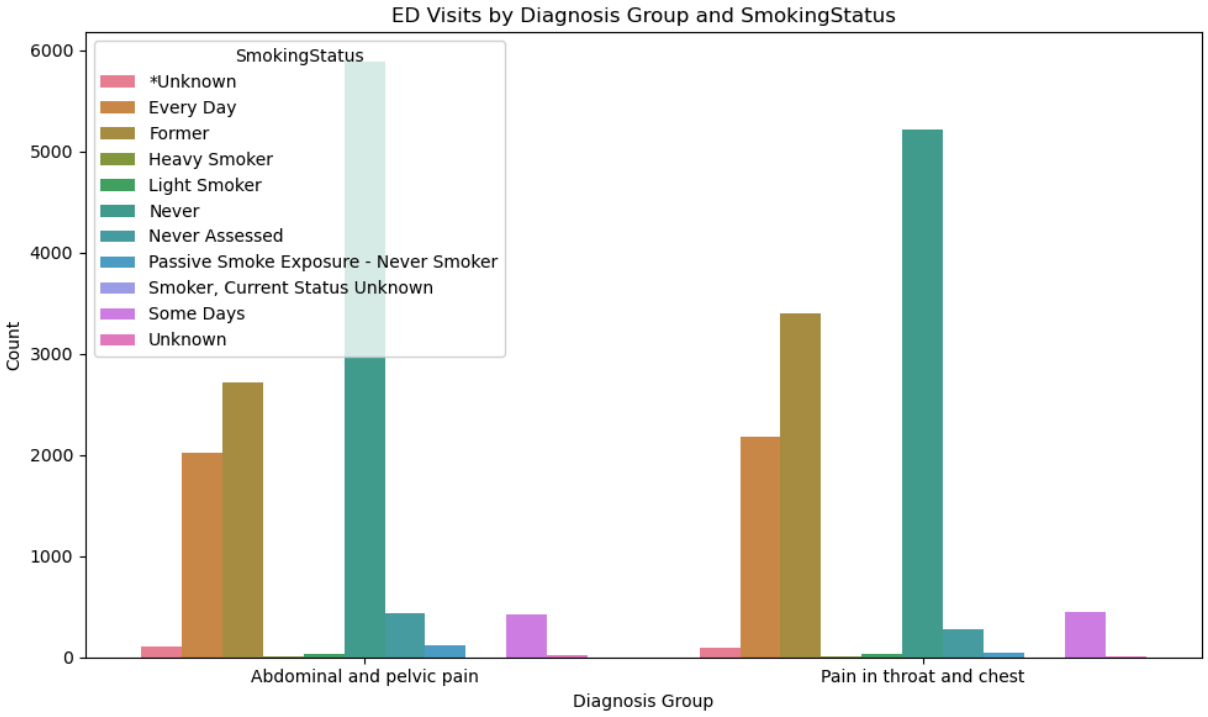
In [29]:
```
grouped_ed = emergency_df.groupby(["GroupName", "FirstRace"]).size().reset_index(na
plt.figure(figsize=(10, 6))
sns.barplot(data=grouped_ed, x="GroupName", y="Count", hue="FirstRace")
plt.title("ED Visits by Diagnosis Group and Race")
plt.xlabel("Diagnosis Group")
plt.ylabel("Count")
plt.xticks(wrap=True)
plt.legend(title="FirstRace")
plt.tight_layout()
plt.show()
```



In [31]:
```
grouped_ed = emergency_df.groupby(["GroupName", "SexAssignedAtBirth"]).size().reset
plt.figure(figsize=(10, 6))
sns.barplot(data=grouped_ed, x="GroupName", y="Count", hue="SexAssignedAtBirth")
plt.title("ED Visits by Diagnosis Group and SexAssignedAtBirth")
plt.xlabel("SexAssignedAtBirth")
plt.ylabel("Count")
plt.xticks(wrap=True)
plt.legend(title="SexAssignedAtBirth")
plt.tight_layout()
plt.show()
```

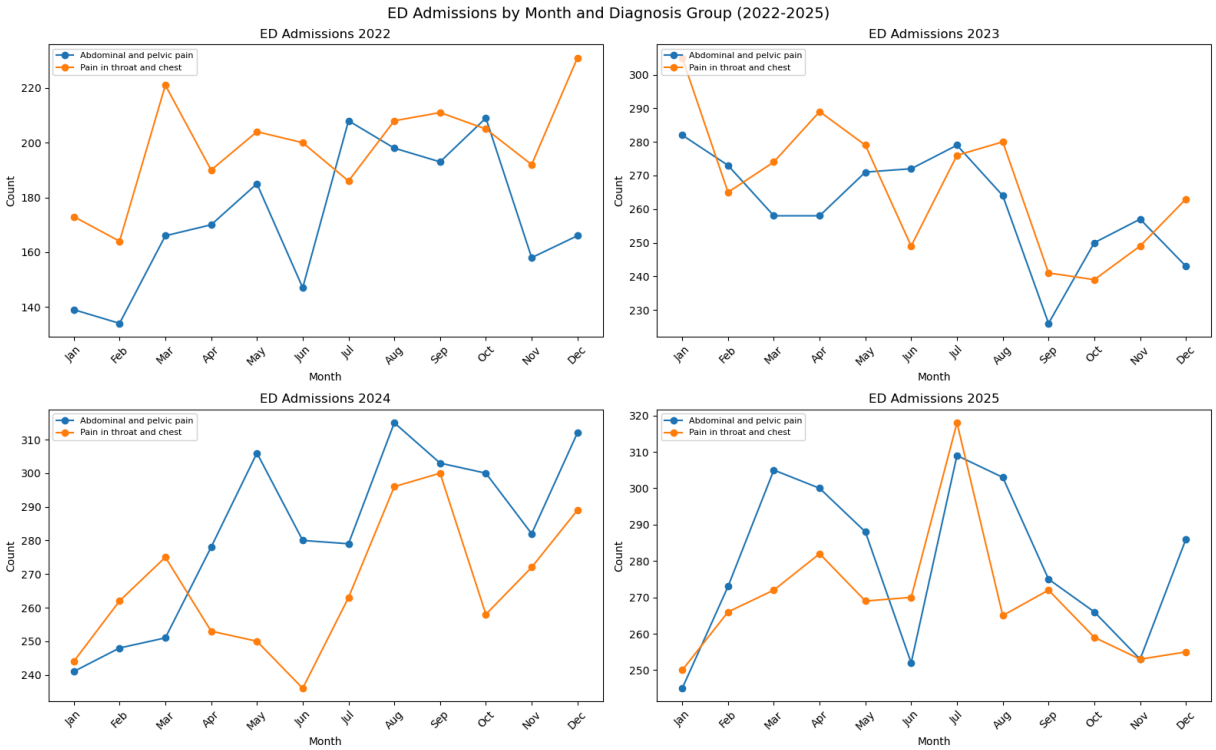


In [32]:
```
grouped_ed = emergency_df.groupby(["GroupName", "SmokingStatus"]).size().reset_inde
plt.figure(figsize=(10, 6))
sns.barplot(data=grouped_ed, x="GroupName", y="Count", hue="SmokingStatus")
plt.title("ED Visits by Diagnosis Group and SmokingStatus")
plt.xlabel("Diagnosis Group")
plt.ylabel("Count")
plt.xticks(wrap=True)
plt.legend(title="SmokingStatus")
plt.tight_layout()
plt.show()
```



In [34]:
```
emergency_df["Month"] = emergency_df["Date"].dt.month
emergency_df["Year"] = emergency_df["Date"].dt.year
emergency_df["YearMonth"] = emergency_df["Date"].dt.to_period("M")

fig, axes = plt.subplots(2, 2, figsize=(16, 10))
axes = axes.flatten()

years = [2022, 2023, 2024, 2025]

for i, year in enumerate(years):
    year_df = emergency_df[emergency_df["Year"] == year]
    grouped = year_df.groupby(["Month", "GroupName"]).size().reset_index(name="Coun
    grouped = grouped.sort_values("Month")

    for group, data in grouped.groupby("GroupName"):
        axes[i].plot(data["Month"], data["Count"], marker='o', label=group)

    axes[i].set_title(f"ED Admissions {year}")
    axes[i].set_xlabel("Month")
    axes[i].set_ylabel("Count")
    axes[i].set_xticks(range(1, 13))
    axes[i].set_xticklabels(['Jan','Feb','Mar','Apr','May','Jun','Jul','Aug','Sep',
```

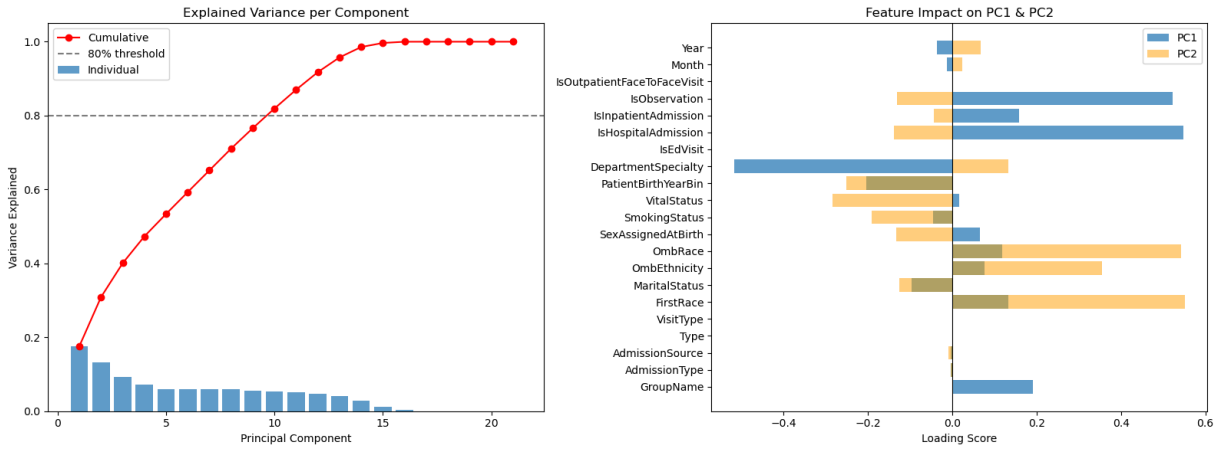
```
    axes[i].legend(loc='upper left', fontsize=8)

plt.suptitle("ED Admissions by Month and Diagnosis Group (2022-2025)", fontsize=14)
plt.tight_layout()
plt.show()
```



From our visualization there from times around june and around fall (October November) the admission counts for these two issues occur the most around these seasons. I think running a *PCA analysis* and understanding what variables highly contribute to theses cases can help us further understand this trend we are observing within the data.

In [36]:
```
pca_df = emergency_df.copy()

cat_cols = [
    'GroupName', 'AdmissionType', 'AdmissionSource', 'Type',
    'VisitType', 'FirstRace', 'MaritalStatus', 'OmbEthnicity',
    'OmbRace', 'SexAssignedAtBirth', 'SmokingStatus',
    'VitalStatus', 'PatientBirthYearBin', 'DepartmentSpecialty'
]

feature_cols = [
    'GroupName', 'AdmissionType', 'AdmissionSource', 'Type',
    'VisitType', 'FirstRace', 'MaritalStatus', 'OmbEthnicity',
    'OmbRace', 'SexAssignedAtBirth', 'SmokingStatus',
    'VitalStatus', 'PatientBirthYearBin', 'DepartmentSpecialty',
    'IsEdVisit', 'IsHospitalAdmission', 'IsInpatientAdmission',
    'IsObservation', 'IsOutpatientFaceToFaceVisit', 'Month', 'Year'
]

label = LabelEncoder()
for col in cat_cols:
    if col in pca_df.columns:
```

```
        pca_df[col] = label.fit_transform(pca_df[col].astype(str))

X = pca_df[feature_cols].fillna(0)
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

pca = PCA()
pca.fit(X_scaled)

fig, axes = plt.subplots(1, 2, figsize=(16, 6))
explained = pca.explained_variance_ratio_
cumulative = np.cumsum(explained)

axes[0].bar(range(1, len(explained)+1), explained, alpha=0.7, label='Individual')
axes[0].plot(range(1, len(explained)+1), cumulative, marker='o', color='red', label
axes[0].axhline(y=0.8, color='gray', linestyle='--', label='80% threshold')
axes[0].set_xlabel('Principal Component')
axes[0].set_ylabel('Variance Explained')
axes[0].set_title('Explained Variance per Component')
axes[0].legend()

loadings = pca.components_[:2].T
axes[1].barh(feature_cols, loadings[:, 0], alpha=0.7, label='PC1')
axes[1].barh(feature_cols, loadings[:, 1], alpha=0.5, label='PC2', color='orange')
axes[1].set_xlabel('Loading Score')
axes[1].set_title('Feature Impact on PC1 & PC2')
axes[1].axvline(x=0, color='black', linewidth=0.8)
axes[1].legend()

plt.tight_layout()
plt.show()
```



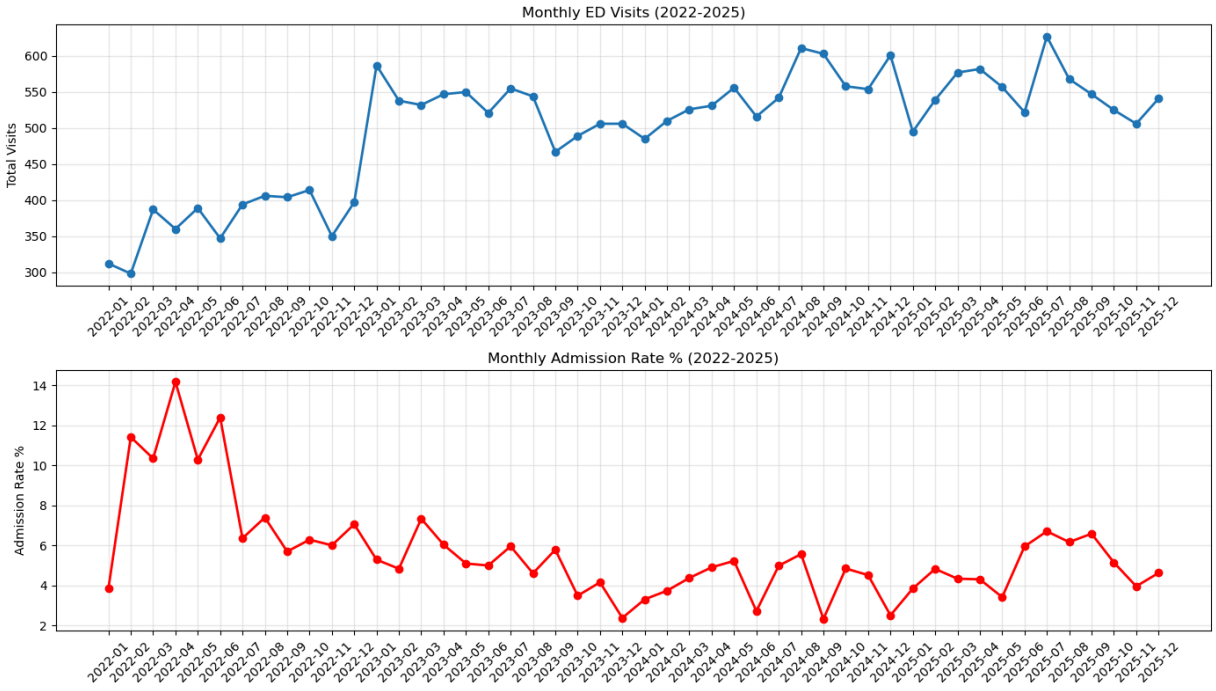
Lets see the admissioin rate for people after going to the ED

In [80]:
```
monthy_admission = duckdb.sql("""
    SELECT
        DATE_TRUNC('month', Date) AS Month,
        COUNT(*) AS Total_Visits,
        SUM(IsHospitalAdmission) AS Total_Admissions
    FROM combined_data
    WHERE IsEdVisit = 1
    GROUP BY DATE_TRUNC('month', Date)
```

```
    ORDER BY Month
""").df()
```

```
FloatProgress(value=0.0, layout=Layout(width='auto'), style=ProgressStyle(bar_color
='black'))
```

In [81]:
```
from statsmodels.tsa.holtwinters import ExponentialSmoothing

fig, axes = plt.subplots(2, 1, figsize=(14, 8))

axes[0].plot(monthly_admissions['YearMonth'], monthly_admissions['Total_Visits'], m
axes[0].set_title('Monthly ED Visits (2022-2025)')
axes[0].set_ylabel('Total Visits')
axes[0].tick_params(axis='x', rotation=45)
axes[0].grid(True, alpha=0.3)

axes[1].plot(monthly_admissions['YearMonth'], monthly_admissions['Admission_Rate'],
axes[1].set_title('Monthly Admission Rate % (2022-2025)')
axes[1].set_ylabel('Admission Rate %')
axes[1].tick_params(axis='x', rotation=45)
axes[1].grid(True, alpha=0.3)

plt.tight_layout()
plt.show()

#Forecast the next 6 months
model_visits = ExponentialSmoothing(
    monthly_admissions['Total_Visits'],
    trend='add',
    seasonal='add',
    seasonal_periods=12
).fit()

model_admissions = ExponentialSmoothing(
    monthly_admissions['Admission_Rate'],
    trend='add',
    seasonal='add',
    seasonal_periods=12
).fit()

forecast_visits = model_visits.forecast(6)
forecast_admissions = model_admissions.forecast(6)

print("=== 6 Month Forecast ===")
for i, (v, a) in enumerate(zip(forecast_visits, forecast_admissions)):
    print(f"Month +{i+1}: {v:.0f} visits | {a:.2f}% admission rate")
```




```
=== 6 Month Forecast ===
Month +1: 563 visits | 3.59% admission rate
Month +2: 564 visits | 6.02% admission rate
Month +3: 598 visits | 6.70% admission rate
Month +4: 598 visits | 7.73% admission rate
Month +5: 606 visits | 6.65% admission rate
Month +6: 569 visits | 7.41% admission rate
```

This is very interesting to see but it makes a lot of sense. During 2022 that was covid era meaning the admission rates to hospitalize patients that had Covid-19 were in the rise at the time. However, though the rate receded oertime, the total amount of people admitted to the hospitals remained the same and did not return to levels before covid

TRAINING RandomForrest & Logistic Regression

Exploring Hosipital Admissions

DESCRIPTION:

In [90]:
```
emergency_df = duckdb.sql("""
    SELECT *
    FROM combined_data
    WHERE IsEdVisit = '1'
""").df()
```

```
FloatProgress(value=0.0, layout=Layout(width='auto'), style=ProgressStyle(bar_color
='black'))
```

In [91]:
```
print(f"Admitted: {(emergency_df['IsHospitalAdmission'] == 1).sum()}")
print(f"Not Admitted:{(emergency_df['IsHospitalAdmission'] == 0).sum()}")
```

```
Admitted: 54135
Not Admitted:207323
```

This isn't good. We have a massive class imbalance where more than 90% of the encounters are either not admited to the hospital. We will be using *SMOTE* package to resample and balance the classifiers

In [92]:
```
drop_cols = [
    # All null
    'ClinicianTitle', 'OfficeAddress', 'OfficeCity', 'OfficePostalCode',
    'PrimaryDepartment', 'PrimarySpecialty',
    # Redundant time
    'AdmissionInstant', 'DischargeInstant',
    'AdmitYear', 'AdmitMonth', 'AdmitDay', 'AdmitHour', 'AdmitMinute',
    'DischargeYear', 'DischargeMonth', 'DischargeDay', 'DischargeHour', 'DischargeM
    # Redundant keys
    'AttendingProviderDurableKey', 'DischargeProviderDurableKey'
]

emergency_df = emergency_df.drop(columns=drop_cols)

emergency_df["Month"] = emergency_df["Date"].dt.month
emergency_df["Year"] = emergency_df["Date"].dt.year
emergency_df["YearMonth"] = emergency_df["Date"].dt.to_period("M")
```

In [93]:
```
full_pca_df = emergency_df.copy()

cat_cols = [
    'GroupName', 'OmbRace', 'FirstRace', 'OmbEthnicity',
    'SexAssignedAtBirth', 'SmokingStatus', 'MaritalStatus',
    'PatientBirthYearBin'
]

le = LabelEncoder()
for col in cat_cols:
    if col in full_pca_df.columns:
        full_pca_df[col] = le.fit_transform(full_pca_df[col].astype(str))

feature_cols = [
    'GroupName', 'OmbRace', 'FirstRace', 'OmbEthnicity',
    'SexAssignedAtBirth', 'SmokingStatus', 'MaritalStatus',
    'PatientBirthYearBin', 'Month', 'Year'
]

X = full_pca_df[feature_cols].fillna(0)
y = full_pca_df['IsHospitalAdmission']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=67, stratify=y
)

smote = SMOTE(random_state=67)
X_train_sm, y_train_sm = smote.fit_resample(X_train, y_train)

print("After SMOTE:")
print(f"Class 0: {sum(y_train_sm == 0)}")
print(f"Class 1: {sum(y_train_sm == 1)}")
```

```
After SMOTE:
Class 0: 165858
Class 1: 165858
```

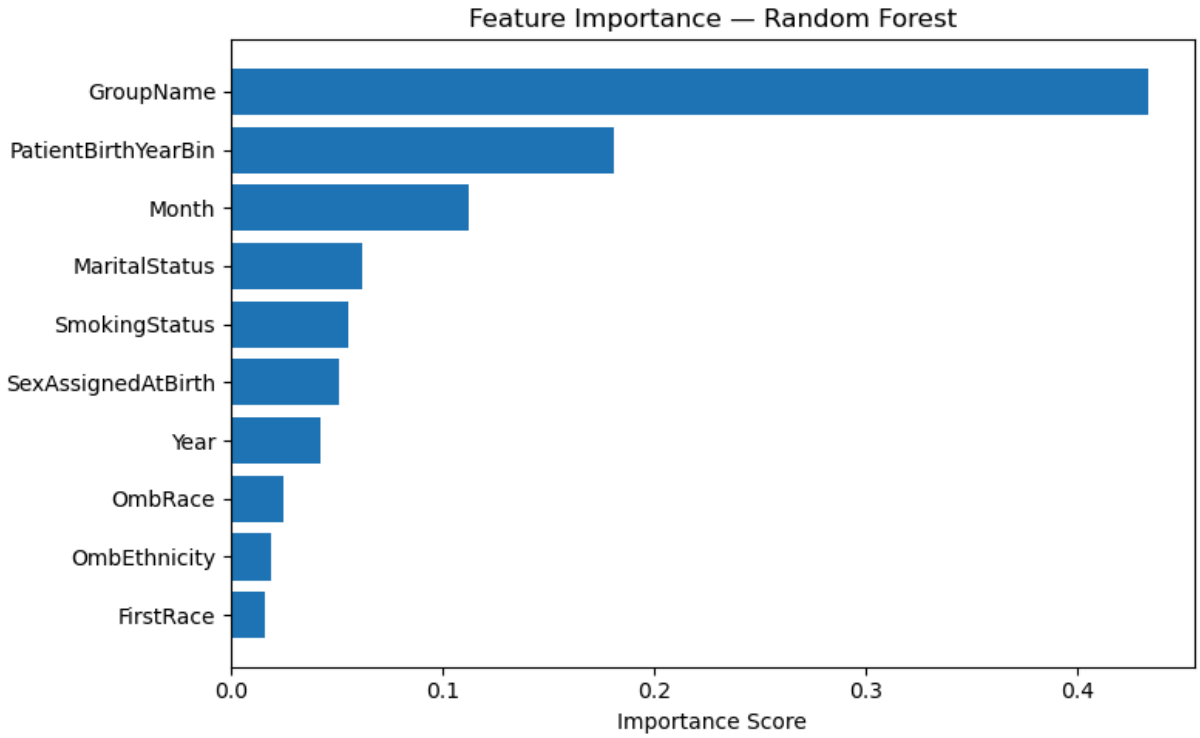
In [94]:
```
lr_model = LogisticRegression(max_iter = 5000, class_weight="balanced", random_stat
lr_cv = cross_val_score(lr_model, X_train_sm, y_train_sm, cv=5, scoring='accuracy')

forest_model = RandomForestClassifier(n_estimators=100, class_weight="balanced", ra
forest_cv = cross_val_score(forest_model, X_train_sm, y_train_sm, scoring="accuracy
```

In [96]:
```
print("--Logistic Regression Classifier--")
print(f"CV Scores: {lr_cv.round(4)}")
print(f"CV Average: {lr_cv.mean().round(4)}")
print(f"CV Standard dev: {lr_cv.std().round(4)}")

print("--RandomForest Classifier--")
print(f"CV Scores: {forest_cv.round(4)}")
print(f"CV Average: {forest_cv.mean().round(4)}")
print(f"CV standard dev: {forest_cv.std().round(4)}")
```

```
--Logistic Regression Classifier--
CV Scores: [0.6799 0.6846 0.6878 0.689  0.6846]
CV Average: 0.6852
CV Standard dev: 0.0032
--RandomForest Classifier--
CV Scores: [0.8156 0.8731 0.8986 0.8972 0.8968]
CV Average: 0.8762
CV standard dev: 0.0318
```

In [97]:
```
forest_model.fit(X_train_sm, y_train_sm)
lr_model.fit(X_train_sm, y_train_sm)

y_pred_rf = forest_model.predict(X_test)
y_pred_lr = lr_model.predict(X_test)

print("-- Random Forest--")
print(classification_report(y_test, y_pred_rf, target_names=['Not Admitted', 'Admit

print("-- Logistic Regression --")
print(classification_report(y_test, y_pred_lr, target_names=['Not Admitted', 'Admit

# --- Feature Importance ---
importance_df = pd.DataFrame({
    'Feature': feature_cols,
    'Importance': forest_model.feature_importances_
}).sort_values('Importance', ascending=False)

plt.figure(figsize=(8, 5))
plt.barh(importance_df['Feature'], importance_df['Importance'])
plt.title('Feature Importance — Random Forest')
plt.xlabel('Importance Score')
plt.gca().invert_yaxis()
plt.tight_layout()
plt.show()
```

```
=== Random Forest ===
              precision    recall  f1-score   support

Not Admitted       0.90      0.87      0.89     41465
    Admitted       0.57      0.64      0.60     10827

    accuracy                           0.82     52292
   macro avg       0.73      0.75      0.74     52292
weighted avg       0.83      0.82      0.83     52292

=== Logistic Regression ===
              precision    recall  f1-score   support

Not Admitted       0.90      0.63      0.74     41465
    Admitted       0.34      0.72      0.46     10827

    accuracy                           0.65     52292
   macro avg       0.62      0.67      0.60     52292
weighted avg       0.78      0.65      0.68     52292
```



In [ ]: